

LOURDES DE BLANC SABLON, QC, Canada

WMO: 718080

Lat: 51.443N

Lon: 57.189W

Elev: 37

StdP: 100.88

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.5	(c) -22.3	(d) -31.3	(e) 0.2	(f) -24.1	(g) -29.0	(h) 0.3	(i) -21.9	(j) 21.3	(k) -5.3	(l) 19.0	(m) -4.9	(n) 5.7	(o) 270	(p) 0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.3	(c) 19.2	(d) 15.4	(e) 17.9	(f) 15.0	(g) 16.7	(h) 14.4	(i) 16.5	(j) 18.0	(k) 15.7	(l) 17.0	(m) 15.0	(n) 16.2	(o) 4.9	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.4	17.0	15.1	10.8	16.2	14.4	10.3	15.7	46.6	18.0	44.3	17.1	42.2	16.3	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	15.3	13.0	-27.5	23.0	2.6	1.6	-29.4	24.2	-30.9	25.1	-32.4	26.0	-34.3	27.2		
			-27.9	18.7	2.5	1.1	-29.7	19.5	-31.2	20.1	-32.6	20.8	-34.4	21.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-10.8	-10.8	-6.5	-1.0	3.9	8.4	12.4	13.7	9.6	4.6	-0.8	-6.5		
	DBStd	9.33	6.43	6.36	5.33	3.39	2.52	2.68	2.12	2.10	2.85	2.96	4.17	5.73		
	HDD10.0	3368	645	582	512	330	191	61	4	2	41	169	323	510		
	HDD18.3	6169	903	815	770	580	448	297	183	144	262	427	573	768		
	CDD10.0	240	0	0	0	0	1	14	80	116	28	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	2	0	0	0	0	0	1	1	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.5	6.5	6.5	6.6	6.0	4.9	4.3	3.8	4.3	5.0	5.5	6.2	6.7		
	PrecAvg	1110	109	92	95	66	69	82	97	107	98	95	90	111		
(15) Precipitation	PrecMax	1510	353	165	177	135	156	182	222	239	204	206	153	175		
	PrecMin	865	49	41	58	37	9	42	38	45	53	47	42	46		
	PrecStd	134	61	32	27	22	34	29	36	46	38	37	29	33		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	4.1	3.8	9.1	15.4	20.3	21.4	21.7	18.6	13.9	10.8	5.3		
		MCWB	3.1	3.2	2.5	6.2	9.9	15.1	16.3	16.7	15.4	11.9	10.1	4.7		
	2%	DB	1.9	2.1	2.3	5.9	11.9	16.9	19.0	19.2	16.6	11.7	7.6	3.7		
		MCWB	1.3	1.2	1.2	4.0	7.8	12.6	15.3	15.7	14.0	10.3	7.2	3.2		
	5%	DB	0.3	0.7	1.4	4.2	9.6	14.6	17.2	18.1	15.3	10.1	6.0	2.6		
		MCWB	-0.2	-0.1	0.4	2.7	6.6	11.3	14.5	15.4	13.2	8.8	5.4	1.9		
	10%	DB	-1.2	-1.0	0.3	3.1	8.0	12.8	16.0	17.2	14.1	9.0	4.6	1.4		
		MCWB	-1.8	-1.8	-0.5	1.7	5.8	10.3	13.8	15.0	12.5	7.7	4.0	0.7		
	0.4%	WB	3.4	3.3	3.0	7.0	11.2	15.6	17.7	18.1	16.5	12.4	10.0	5.0		
		MCDB	3.7	3.6	3.6	8.9	13.7	19.2	19.9	19.9	17.5	13.3	10.7	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.3	1.5	1.5	4.3	8.7	13.5	16.1	16.7	14.9	10.6	7.3	3.3		
		MCDB	1.8	2.1	2.0	5.3	11.1	15.8	17.8	18.1	16.1	11.4	7.5	3.6		
	5%	WB	-0.1	-0.1	0.6	2.9	7.1	11.9	15.1	16.1	13.8	9.2	5.6	2.0		
		MCDB	0.3	0.6	1.2	3.9	9.0	13.8	16.5	17.3	14.8	9.9	5.9	2.5		
	10%	WB	-1.8	-1.7	-0.4	1.9	6.0	10.7	14.3	15.5	12.8	8.0	4.1	0.8		
		MCDB	-1.3	-1.0	0.3	2.9	7.6	12.3	15.5	16.6	13.9	8.8	4.5	1.3		
	5% DB	MDBR	8.2	8.0	7.1	5.6	6.9	7.1	6.3	6.3	6.6	5.7	5.8	6.9		
		MCDBR	8.2	6.8	5.7	6.4	9.9	10.5	8.6	7.6	7.5	6.7	6.2	6.2		
		MCWBR	8.3	6.5	5.3	5.1	6.8	6.7	5.5	5.1	5.7	5.8	6.2	6.3		
		MCWBR	8.4	7.4	5.8	6.0	8.8	9.0	7.3	6.0	6.5	6.4	6.3	6.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.6	7.2	5.6	5.0	6.6	6.4	5.2	4.5	5.6	5.9	6.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.280	0.303	0.344	0.377	0.377	0.410	0.377	0.351	0.320	0.290	0.275		
	taud		2.319	2.329	2.298	2.280	2.330	2.383	2.345	2.435	2.478	2.524	2.501	2.362		
	Ebn at Noon		744	853	897	894	877	878	840	853	838	802	740	687		
	Edh at Noon		63	82	103	119	119	114	117	102	87	69	54	52		
(44) All-Sky Solar Radiation	RadAvg		0.85	1.47	2.52	3.73	4.89	5.06	4.91	4.19	3.08	1.84	1.01	0.67		
	RadStd		0.12	0.26	0.36	0.58	0.40	0.46	0.37	0.30	0.21	0.14	0.10	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	+1.74	N/A	+0.35	+0.48	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	+1.67	N/A	+0.38	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page